

Igor Lucindo Cardoso

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Lubbock, TX

OBJECTIVE

PhD student specializing in Operations Research, with a focus on large-scale optimization. Interested in integrating expertise in engineering and mathematical modeling to tackle complex real-world problems.

EDUCATION

Texas Tech University (TTU) 2025 – Present
Ph.D. in Industrial Engineering
GPA: 4.00/4.00
• *Research interests:* Optimization, Mixed-Integer Programming (MIP), Graph Theory.

Getulio Vargas Foundation (FGV) 2023 – 2024
M.A. Courses in Mathematical Modeling
• *Relevant coursework:* Measure Theory, Probability Theory.

Military Institute of Engineering (IME) 2020 – 2024
B.S. in Electronic Engineering
GPA: 8.25/10.00
• IME is the oldest and one of the best-ranked engineering schools in Brazil.

PUBLICATIONS

- **Cardoso, I.,** Teodoro, G., Validi, H. (2026). **A Polytime and Interpretable Approach for Solving Wordle.** Submitted to *Operations Research*.
- Gutierrez, R., **Cardoso, I.,** Validi, H. (2026). **Sport Scheduling to Minimize Travels at the FIFA World Cup 2026.** Submitted to *Optimization Letters*.
- **Cardoso, I.,** et al. (2024). **Comparing Pre-Trained Object Detection Models for Autonomous Grasp on Affordable Prosthetic Hands.** *IEEE MeMeA*.
- *Manuscripts in Progress:* **Cardoso, I.** and Validi, H. **On Solving the Network-Based Portfolio Optimization Problems.** *In prep*, 2026.

RESEARCH EXPERIENCE

Research Assistant, Texas Tech University (TTU) 2025 – Present
• Conducting doctoral research in Operations Research, focusing on optimization, MIP, and graph theory.
• Developing novel algorithms for NP-hard combinatorial problems, leveraging GPU acceleration and decomposition techniques to solve large-scale instances. *Advisor: Dr. Hamidreza Validi.*

Research Experience for Undergraduates, Memorial University (MUN) 2023
• Developed autonomous prosthetic grasping capabilities, optimizing machine learning and computer vision inference pipelines to achieve real-time predictive control. *Advisor: Dr. Vinicius P. da Fonseca.*

HONORS & AWARDS

- TTU Tye Industrial Endowment Engineering Scholarship, Texas Tech University 2026
- Student Chapter Annual Award (Cum Laude), INFORMS 2025
- 1st place in "The Road to 2050" competition, Shell Eco-marathon Team 2021
- 5th place in Petrobras Challenge, Latin American Robotics Competition, RoboIME 2021
- Gold and Silver Medals, Brazilian Mathematical Olympiad of Public Schools (OBMEP) 2018 – 2019

EXTRACURRICULAR EXPERIENCE

INFORMS Student Chapter, Texas Tech University 2025 – Present
• Active member engaging with the research community in Operations Research.

Journal Club, Texas Tech University 2025 – Present
• Presenter in Journal Club sessions, analyzing peer-reviewed research in Operations Research.

American Airlines Scheduling Project 2025
• Developed mixed-integer programming models to optimize aircraft scheduling for American Airlines.

Robotics Club - RoboIME 2021 – 2023
• Contributed to projects involving machine learning and computer vision at the Military Institute of Engineering.

SOFTWARE & PROJECTS

Wordle Optimization Solver Python, CuPy, CUDA
Leveraged GPU-accelerated array operations for highly parallelized execution. [\[View App\]](#)

FIFA World Cup 2026 Scheduler

Interactive web app using MIP to optimize match scheduling and minimize travel distance. [\[View App\]](#)

Python, JS, MIP Solvers

Expense Splitter

Optimization tool to minimize the number of transactions required to settle shared expenses. [\[View App\]](#)

Python, Linear Programming

Computer Vision Rubik's Cube Solver

Built a real-time web app mapping physical states to a 3D digital twin to generate CFOP solutions. [\[View App\]](#)

JS, OpenCV

SKILLS

Programming: C/C++, Python, Matlab, Julia, Java, JavaScript, HTML, CSS, FPGA, SQL

Tools: Gurobi, Cuda/GPU, Git, $\LaTeX$, Linux, NetworkX, ROS, OpenCV, Arduino, Cloud Firestore

Languages: Portuguese (native), English (fluent), French (intermediate)

Coursework: Network Optimization, Measure Theory, Probability Theory, Machine Learning, Algorithms, Computer Vision, Computer Architecture.

REFERENCES



Dr. Hamidreza Validi (advisor) - hvalidi@ttu.edu



Dr. Burak Eksioglu (professor) - burak.eksioglu@ttu.edu



Dr. Ningji Wei (professor) - ningji.wei@ttu.edu



Dr. Vinicius P. da Fonseca (advisor) - vpradodafons@mun.ca